Kyle Tam and Andre Smith

GitHub: https://github.com/kyletam1/sleep-analysis-project

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
# Import Packages

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestRegressor
```

Problem Statement

This project examines how demographic, behavioral, and health related factors such as age, physical activity, stress level, and BMI, affect sleep duration and quality. The objective is to identify which variables are most impactful in order to better understand patterns associated with healthy sleep.

```
SleepHealth = pd.read_csv('/content/drive/My Drive/Colab Notebooks/ss.csv')
SleepHealth.head()
```

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	1	Male	27	Software Engineer	6.1	6	42	6	Overweight	126/83	77	4200	NaN
1	2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
2	3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
4	5	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea

Next steps: Generate code with SleepHealth New interactive sheet

```
# Remove Duplicates

SleepHealth = SleepHealth.drop(columns=['Person ID'])
SleepHealth = SleepHealth.drop_duplicates()
```

```
# Clean Data

SleepHealth['Sleep Disorder'] = SleepHealth['Sleep Disorder'].fillna("None")
SleepHealth.head()
```

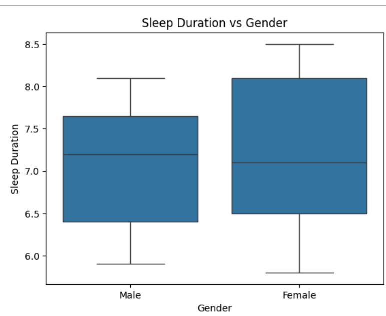
	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	Male	27	Software Engineer	6.1	6	42	6	Overweight	126/83	77	4200	None
1	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	None
3	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
5	Male	28	Software Engineer	5.9	4	30	8	Obese	140/90	85	3000	Insomnia
6	Male	29	Teacher	6.3	6	40	7	Obese	140/90	82	3500	Insomnia

Next steps: Generate code with SleepHealth New interactive sheet

```
# Variables / Data type

print(" #   Column                    Non-Null Count  Dtype  ")
print("---  ------                    --------------  -----  ")

for i, col in enumerate(SleepHealth.columns):
    non_null = SleepHealth[col].notnull().sum()
    dtype = SleepHealth[col].dtype

    print(f" {i:<3} {col:<25}{str(non_null) + ' non-null':<16}{dtype}")
```

```
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Gender                    132 non-null    object
 1   Age                       132 non-null    int64
 2   Occupation                132 non-null    object
 3   Sleep Duration            132 non-null    float64
 4   Quality of Sleep          132 non-null    int64
 5   Physical Activity Level   132 non-null    int64
 6   Stress Level              132 non-null    int64
 7   BMI Category              132 non-null    object
 8   Blood Pressure            132 non-null    object
 9   Heart Rate                132 non-null    int64
 10  Daily Steps               132 non-null    int64
 11  Sleep Disorder            132 non-null    object
```

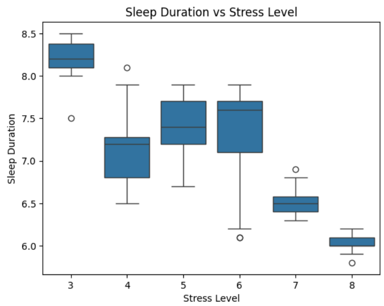
```
# Sleep Duration vs Gender

sns.boxplot(data=SleepHealth, x='Gender', y='Sleep Duration')
plt.title("Sleep Duration vs Gender")
plt.show()
```



Observation

We observed that there was little difference in sleep duration between the two genders. They have similar median values, although the spread appears to be slightly higher for females. Overall, we concluded that gender has a weak impact on sleep duration.

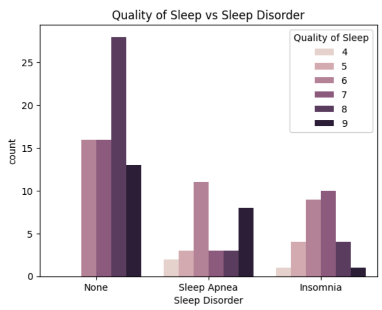
```
# Sleep Duration vs Stress Levels

sns.boxplot(data=SleepHealth, x='Stress Level', y='Sleep Duration')
plt.title("Sleep Duration vs Stress Level")
plt.show()
```



Observation

We observed a clear trend in this variable, as stress levels increased, sleep duration tended to decrease. However, we did notice an anomaly at stress level 4, which could be due to other factors and is worth further investigation. Overall, we concluded that stress level has a strong impact on sleep duration.

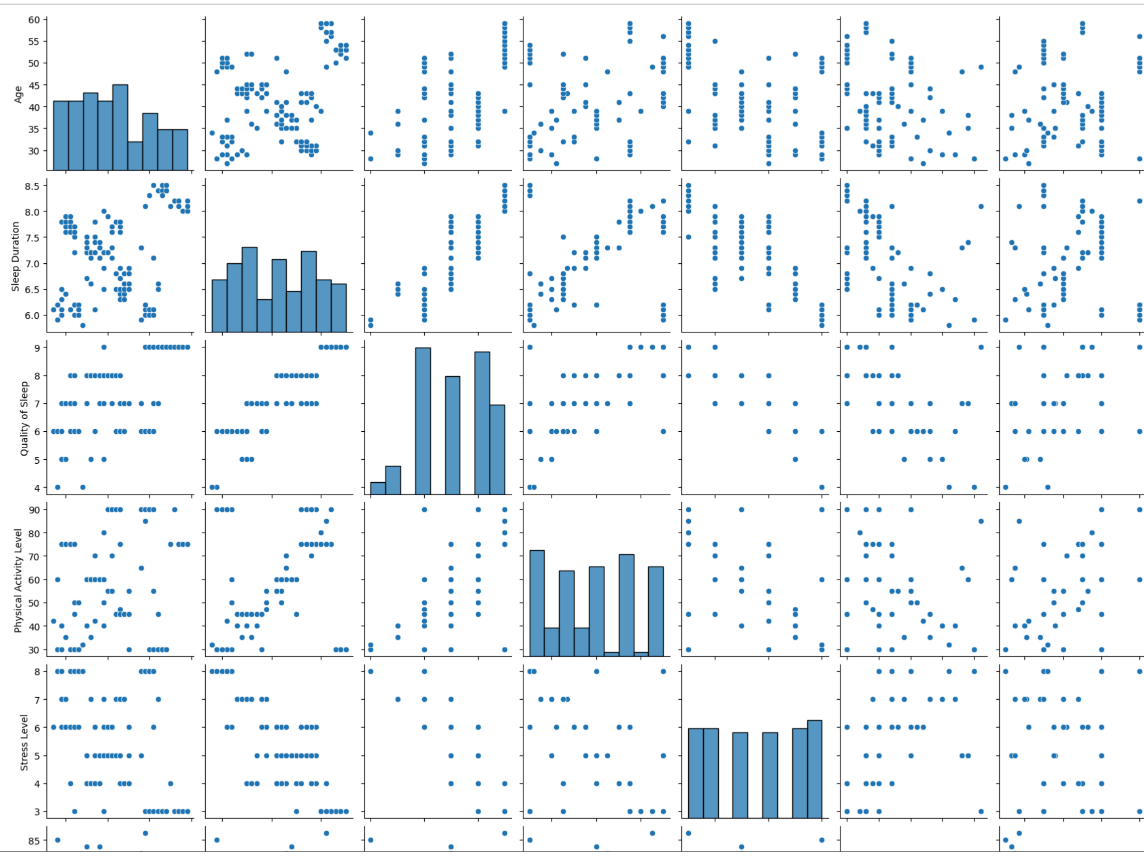
```
# Sleep Quality vs Sleep Disorder

sns.countplot(data=SleepHealth, x='Sleep Disorder', hue='Quality of Sleep')
plt.title("Quality of Sleep vs Sleep Disorder")
plt.show()
```



Observation

We see a clear inference of this data here. Starting with no sleeping disorder, we noticed a lot of them rated sleep quality higher, with 8 being the most common. We interpreted that as they had a good quality of sleep. Next we have those with Sleep Apnea, which is problem with breathing during sleep. We noticed a larger spread in answers with 6 and 9 as the most common. We interpreted that as some either notice the sleeping disorder or they don't and depending on that that affects what answer they put. Last we has isomnia, which is a dificulty falling asleep. We noticed the same spread as Sleep Apnea but the most common was 7. We expected it to be lower but it does make some sense because you can't rate the quality of sleep if you are not sleeping and when you are sleeping it would just follow as those without a sleeping disorder. So overall we said that sleeping disorder has a moderate impact on sleep quality.

```
# Explore relationships between variables

sns.pairplot(SleepHealth.iloc[:,:12])
plt.show()
```



Double-click (or enter) to edit

Double-click (or enter) to edit

Double-click (or enter) to edit

Here we added all the different scatterplots for the different numerical values. Based on this we can see more clearly which variables have a strong relationship to observe. We plan on going more in depth on some of them, just as we have above.

Results

The conclusion we came to so far with the relationships we explored is that stress has a large impact on the duration of someone's sleep. And from the array of scatterplots above we expect to see a strong relationship with stress levels vs sleep duration and quality of sleep. Additionally, sleep disorders appear to have a moderate impact on sleep quality.

Future Work

We wanted to work a bit more on incorpating some more of the categorical data as those could have a better relationship within that. We also wanted to work with visualizing more than just two variables at once. Also, explore how combinations of variables affect sleep.

Kyle: Worked on data cleaning, coding, and visualization

Andre: Contributed to problem formulation, analysis and presentation of results

Both team members collaborated on interpreting results and writing the notebook